Longitudinal analysis of cross country results

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Abstract

This report conducts a longitudinal analysis on the results of the boys and girls cross country teams at Leland High School to determine if there is significant variation between years, and if other factors like weather have an effect. This can aid in possibly predicting future performance of the team. The results show no significant difference in both teams across years. However, the athlete's grade level and their ability to improve do seem to have an effect on their results. Ultimately, any patterns in the results are most likely better explained by other variables not used in this analysis, such as an athlete's physical statistics.

1 Introduction

Sports teams are known for having fluctuations in performance over the years, which can be even be grouped into particular "eras".[9] While part of it can be attributed to random variance, some of it is likely related to observable factors. The understanding that develops between members of the same team over time [1][6], pivotal leadership such as Coach K's legendary reign over the Duke basketball team [7], and perhaps even socioeconomic conditions can all affect the team's performance. Because of this, it can be helpful from a coach's perspective to be able to understand and possibly even predict these variations in performance. While teams that compete in football or basketball on the national level may be more well known, this project's focus is the cross country team at Leland High School in San Jose, CA due to prior experience and knowledge.

The Leland cross country team is a co-ed sports team that is active from August to November each year. The team participates in competitions ("meets") with other schools on the weekdays and more casual "invitationals" on the weekends in September and October, as well as a championship meet to determine who advances to higher-level competition in early November. Since 2002, the team has consistently recorded yearly results and held all their meets at the course Montgomery Hill (2.74 Miles), located in San Jose, as well as having the championship meet and occasional invitational at Crystal Springs (2.95 Miles), located in Belmont, CA. The one exception to this is the year 2020 having no results on these courses due to the COVID-19 pandemic. During this time, the team has also had the same head coach. Due to this consistency, I decided to focus on the years starting from 2002 up to the most recent season, 2025.

The girls and boys teams race separately, and both are further divided into varsity, junior varsity (JV), and reserve teams. The varsity and JV teams are the ones that compete for points and are also restricted to seven people per team, although the exact lineup often varies across meets. Points are scored based on the place that the first five members of the team get; to win, as many team members should finish the race as fast as possible[3]. Since athletes on opposing teams do not generally interfere with each other directly, winning does not require intensive strategy beyond being fast, so cross country might not be considered a "team sport" in the traditional sense[5]. However, members of a team can somewhat influence each other by keeping pace with each other, and having a fast group of individuals can set higher standards for the rest of the team. Because of this, it can be more effective to look at the team's performance as a whole rather than splitting it into the different varsity levels, and there is cause to suspect that team performance may wax and wane each year as old cohorts leave and new ones emerge.

2 Methods

2.1 Data Collection

The data for this project was scraped from athletic.net [8], a publicly available website where schools will post their cross country results. A currently existing Python scraper on GitHub made specifically for this website was used and slightly modified for the data collection, which uses Selenium and BeautifulSoup to parse the HTML on the page [4].

The weather data for each course consisted of the temperature in degrees Fahrenheit and relative humidity. This was collected in Python using the meteostat module, which gets data from weather stations closest to a particular geographical point specified by longitude and latitude. For each race date, temperature and humidity data is taken from the closest weather station with both available. The exception to this rule is for Crystal Springs, as San Francisco and Half Moon Bay are weather stations that are close by, but as they are known for being significantly cooler than other regions of the Bay Area [2], data from them was not used.

2.2 EDA

In the data, a **result** is the response variable of time in minutes, and is indexed at the (athlete, date, distance) level. The smaller a result is, the faster the athlete completed the race. Besides the response, the data includes the predictors of athlete name, the date of a result, the grade of the athlete when they achieved a result, the distance of a race in either meters or miles, and whether each result of a specific athlete was their best time ever achieved (PR), best time in that season (SR), or improvement over the last race at the same distance. The observational units for this project are at the athlete level. Additionally, only athletes who had results at either the 2.74 or 2.95 mile distances were included in the dataset. To consolidate the effects of athlete improvement over time, a **"percent improvement"** variable was defined as $100 * [(cumulative_mean(time)) - current_time)/cumulative_mean(time)]$ per distance for each athlete and set to 0 for the athlete's first ever race at that distance. Summary statistics for both teams were calculated for the predictors and response, and graphs of the relationships between the main predictors and the response were made.

2.3 Model

As the data consisted of repeated observations from the same athletes and was organized in a athlete $\rightarrow$ distance $\rightarrow$ result structure, a linear mixed effects longitudinal hierarchical model was used:

$$time_{ijk} = \beta_0 + \beta_1 grade_{ijk} + \beta_2 improvement_{ijk} + \beta_3 temp_{jk} + \beta_4 humidity_{jk} + \beta_5 year + \beta_6 distance_j + athlete_i + \epsilon_{ijk}$$

The outcome of time in minutes, $time_{ijk}$, is indexed by the athlete i at distance j and date of the result k. The model uses fixed effects for grade, percent improvement, temperature, relative humidity, the year of the race, and the race distance. A random intercept is used for athlete.
Using the lme4 package in R, the girls and boys teams were modeled separately with the same model as their times are obviously going to be different and the difference between them is not of importance. Additionally, the other variables might have interactions with gender, making the model more complicated.

2.3.1 Model Assumptions

The assumptions of this model are that the relationship between the predictors and the response is linear, the random intercept $b_i \sim^{iid} N(0, \tau^2)$ and is independent of both the random errors ϵ_{ijk} and fixed effects, and that the random errors are independent, are centered around zero with constant variance and are normally distributed for inference. In order to assess the model assumptions and performance, the significance of the coefficients, as well as plots of residuals versus fitted values and normal Q-Q plots for the residuals will be checked.

3 Results

The summary statistics are displayed in Figure 9. The median time for boys and girls at Montgomery Hill (2.74 Miles) are 20.27 and 23.72 minutes respectively, with Crystal Springs (2.95 Miles) being very similar at 20.02 and 23.36 minutes. Additionally, temperature and humidity both tend to be higher at Montgomery Hill than Crystal Springs. Besides the overall climate, this is most likely due to both courses typically being ran at different times of the day and year. Examining the predictors of grade, temperature, humidity, and percent improvement while separating by distance:

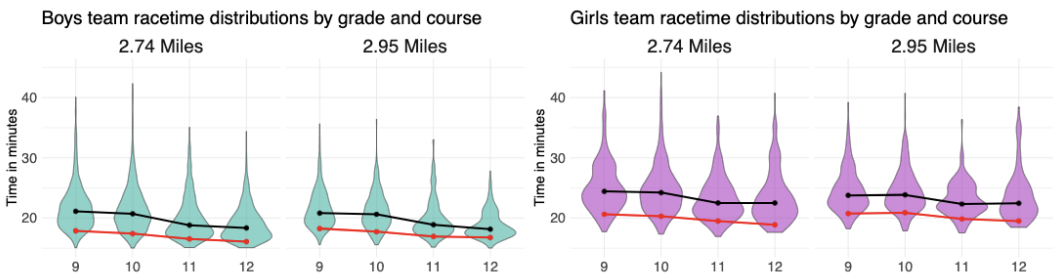

Figure 1: Times tend to improve as athletes get older. Lines at top 50th (black) and 10th (red) percentiles.

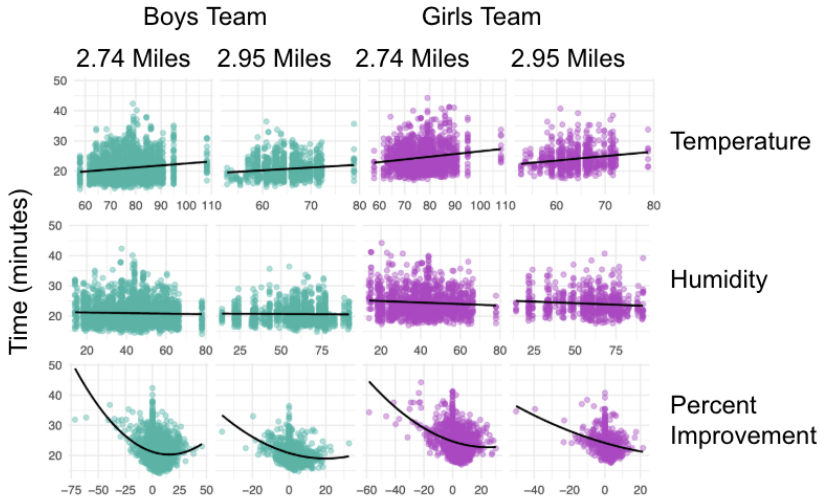

Figure 2: Quantitative predictors temperature, humidity, and percent improvement compared to time.

In Figure 1, there appears to be a noticeable trend of times getting faster as grade increases. This is intuitive, as athletes that are older likely have more experience and are physically stronger. Figure 2 shows higher temperature and lower humidity slightly corresponding to slower times, while the percent improvement appears to exhibit a somewhat non-linear relationship with times, with increasing magnitude in both directions as times get slower. This can be understood as the fact that athletes who are slower have more room to both increase and decrease their times than athletes who are faster, with the fastest athletes having around a 5% rate of improvement over their previous cumulative mean time.

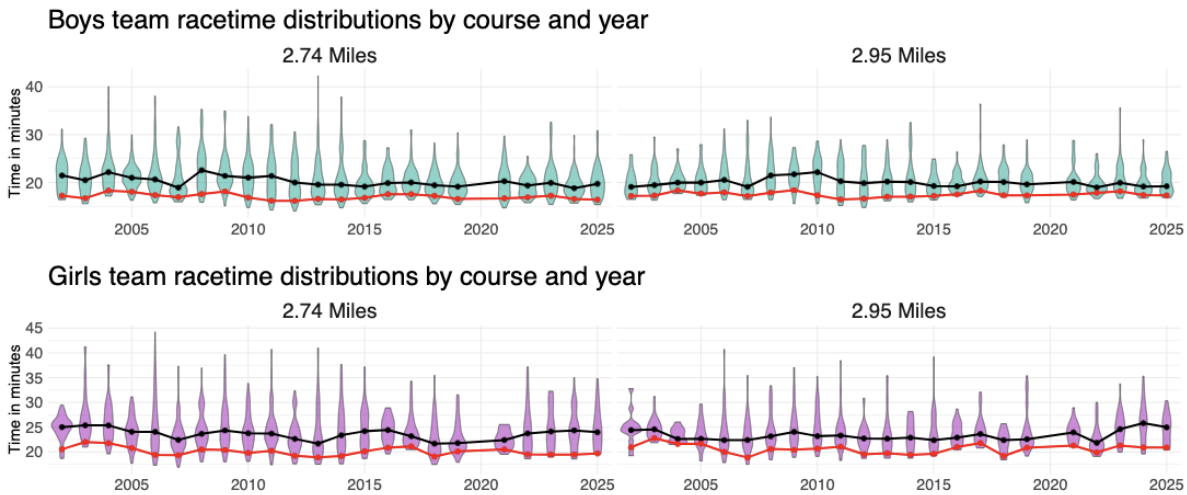

Figure 3: Slight variation in times across seasons, most notably for the girls' team at 2.74 Miles.

In Figure 3, both teams actually have relatively constant performance over time. Besides this, the data is quite skewed, with extremely long right tails (Figure 7, appendix). Keeping this in mind, cluster-robust standard errors were used to quantify the uncertainty of the fixed effects estimates (clubSandwich package in R).

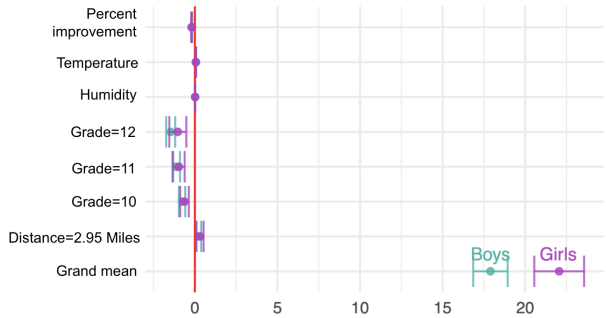

Figure 4: Fixed effects for non-year variables (95% confidence).

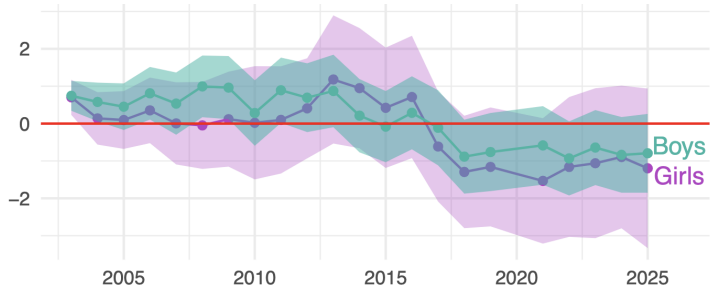

Figure 5: Fixed effects for year, with 2002 as baseline (95% confidence).

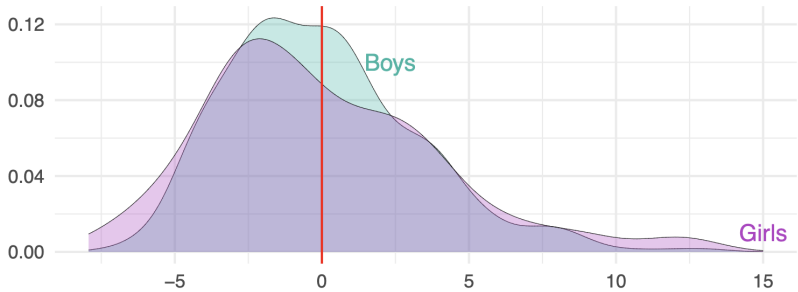

Figure 6: Distribution of the random intercepts for athlete.

The model coefficient estimates for both teams can be found in Tables 1 and 2 in the appendix (6.3). The fixed effects for percent improvement, temperature, and humidity, while highly significant, are close to 0 (Figure 4). For percent improvement, despite the possible non-linear relationship shown in the EDA section, accounting for this by modeling it with a squared term resulted in an even smaller coefficient, so it was kept at a linear level.

None of the fixed effects by year appear to be significant at the 0.05 level, aside from the couple years right after 2002. Observing the confidence interval plots in Figure 4 and 5, the fixed effects for the years on the girls team has a wider 95% confidence interval than the boys, with the width also increasing the farther one gets from the baseline of 2002. Both teams exhibit similar trends in performance, with there appearing to be a shift of the teams tending to be faster rather than slower compared to 2002, starting around 2017.

The random intercepts for athlete have a good amount of variance, with the girls team also appearing to have a larger variance than the boys (Figure 6).

As seen in the residual plots (Figure 10) and Q-Q plots (Figure 11), the model residuals for both teams are heteroskedastic, with the variance tending to increase as the predicted time in minutes increases, and the data also exhibits non-normality with heavy tails, especially on the right side. This and the heavy skewness of the data can be explained by the fact that the team does not typically make cuts based on performance, and athletes on the reserve team are not under pressure to score points. To address these violations, other modeling approaches were attempted, including taking the natural logarithm of the times, performing a Box-Cox transformation, and using a generalized linear mixed effects model with a Gamma distribution. They were ultimately scrapped as they did not fully resolve these issues and were more difficult to interpret, although the Box-Cox transformation did get close for the boys team (Figure 12, 13).

4 Discussion

With the estimated fixed effects for temperature and humidity being 0.05 and 0.01 respectively, for every 1 unit increase in temperature or humidity, holding all other variables constant, times increase by 3 seconds for temperature or 0.6 seconds for humidity on average. Thus, higher temperatures and more humid weather are slightly associated with slower times. This does make sense, as it is harder to run when the weather is more extreme. Increasing grade is associated with faster times, with being in grades

10-12 ranging from around 40 seconds to a full minute of an average drop in times compared to being in grade 9. A one unit increase in percent improvement also causes an average drop in times of around 10 seconds for boys and 12 seconds for girls. The grand intercept of the model (β_0) represents the predicted mean time of an athlete in grade 9 at Montgomery Hill in 2002 with no improvement from past times and if temperature and humidity are somehow zero. Thus the boys team (17.89) has a much faster predicted overall mean time than the girls team (22.06), which is expected. Aside from the overall mean and grade having a slightly larger effect on decreasing boys' times compared to girls', the values of the fixed effects do not differ that much by gender.

The girls having more variance in their coefficient estimates is likely because the boys team has more results and less variance in their times generally (Figure 9), resulting in more confident predictions. This is especially apparent after 2020, where both teams plummet in membership due to the cancellation of activities, but the boys team is able to recover to pre-pandemic numbers while the girls team is not (Figure 8).

4.1 Limitations and Future Work

The data collected is purely observational, thus making it difficult to determine any causal relationships. In addition, the data has a few other restrictions. The weather data is not exact to the time or location of a race, instead being averaged across an approximate window of time when races would take place and being drawn from a nearby weather station. The data only includes results and athletes at two distances; even though these can be argued to make up the majority, they still leave out some results and athletes. Finally, this data is specific to this school and may not be generalizable to other schools.

In terms of future directions, statistics like age, height, and weight of each athlete, which may be more strongly associated with the times, are not available on athletic.net, but could be used to answer the particularly intriguing question of predicting how much a particular athlete will improve over the course of their career. Although several different models were tested due to the non-normality of the data, it is possible there is another distribution that is a better fit.

5 Conclusion

In conclusion, there is no significant difference in results at each course across time, while an increase in grade decreases times and increases in temperature and humidity have a small effect on increasing times. The failure of the model assumptions is most likely due to the fact that the data is very non-normal and other possibly stronger predictors were not available for use in the model.

6 Appendix

6.1 AI Disclosure Statement

AI was used for portions of the coding for this project, mostly to edit graphs for display, and to provide feedback on the modeling direction. Specifically, ChatGPT version 5.2 was used. Full transcripts of the interactions can be found at these links: `https://chatgpt.com/share/6994fcfd-0ec8-8005-8d1d-01633895339b` `https://chatgpt.com/share/69a3b0e4-3ad4-8005-9983-d06110352abe`

6.2 EDA

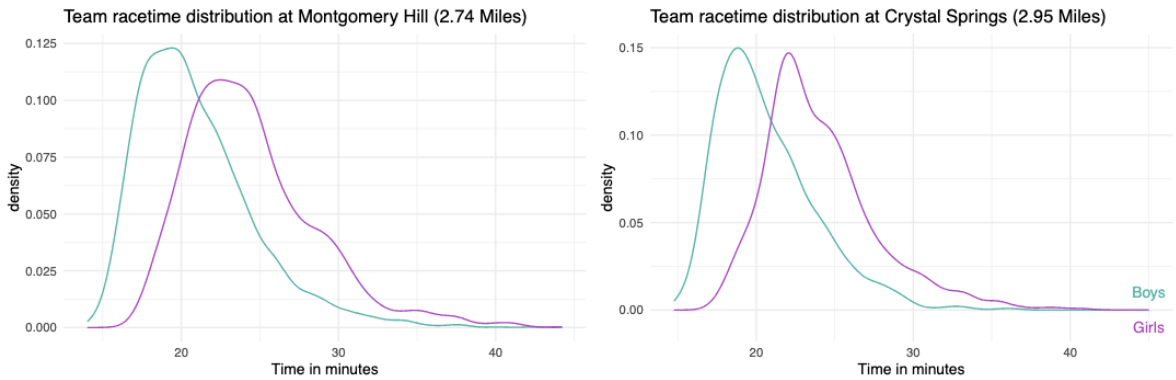

Figure 7: Clear differences in the distribution of times between the boys and girls teams. Also showing the data being significantly skewed right and non-normal.

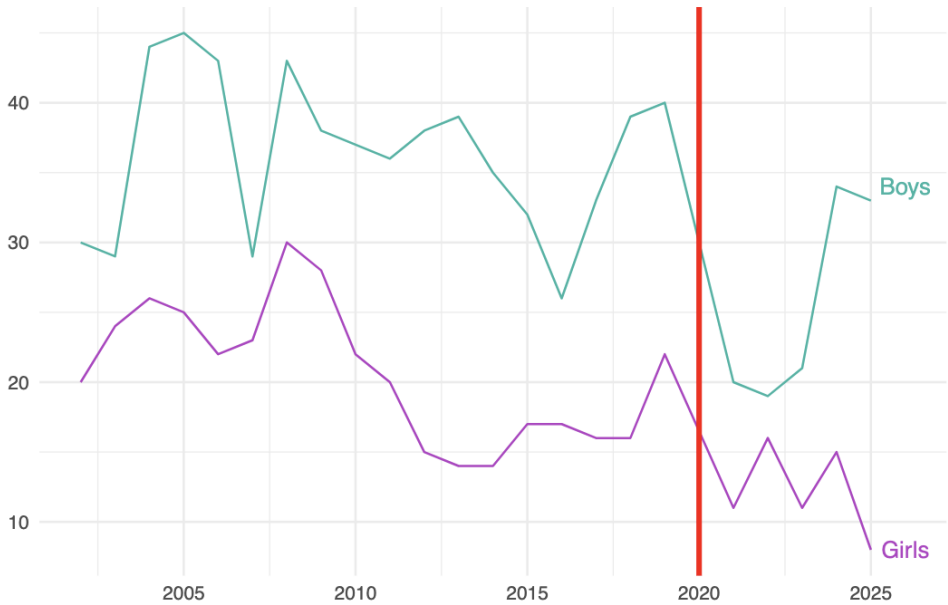

Figure 8: The boys team always has had more members than the girls. Interestingly, both teams saw a significant drop in size after COVID-19, but the boys team was able to recover by 2024, while the girls team did not.

	2.74 Miles		2.95 Miles	
Characteristic	Boys	Girls	Boys	Girls
Number of results	2821	1503	1097	661
Grade: 9	1000 (35.45%)	472 (31.40%)	371 (33.82%)	210 (31.77%)
Grade: 10	995 (35.27%)	531 (35.33%)	348 (31.72%)	218 (32.98%)
Grade: 11	462 (16.38%)	306 (20.36%)	222 (20.24%)	134 (20.27%)
Grade: 12	364 (12.90%)	194 (12.91%)	156 (14.22%)	99 (14.98%)
Time in minutes (Median (IQR))	20.27 (18.23–22.85)	23.72 (21.46–26.60)	20.02 (18.34–22.41)	23.36 (21.71–25.80)
% results that were PR's	16.31%	16.03%	32.91%	27.84%
Grade: 9	16.00%	19.70%	35.85%	29.52%
Grade: 10	18.09%	16.57%	37.64%	33.94%
Grade: 11	12.12%	12.42%	22.52%	23.13%
Grade: 12	17.58%	11.34%	30.13%	17.17%
Temperature by course (F) (Median (IQR))	75.97 (70.70–81.58)		63.74 (60.80–66.76)	
Humidity by course (%) (Median (IQR))	42.00 (32.92–50.00)		61.40 (53.15–73.45)	

Figure 9: Summary statistics of some of the predictors, also including PR rates per grade. The temperature and relative humidity both tend to be higher at the 2.74 mile course than the 2.95 mile course. It appears comparatively easier to PR at 2.95 miles, but this is most likely due to the teams racing less often at this distance.

6.3 Model Coefficients

All numbers in the tables below are rounded to the nearest 4 places.

Table 1: Model coefficient estimates for boys team

	Estimate	CRSE	Degrees of freedom	P-value
Intercept	17.8963	0.5269	124.7106	0.0000
Grade=10	-0.7759	0.0969	149.1867	0.0000
Grade=11	-1.1342	0.1166	134.6480	0.0000
Grade=12	-1.4662	0.1378	119.9534	0.0000
Percent improvement	-0.1796	0.0050	106.3760	0.0000
Temperature	0.0562	0.0049	226.4435	0.0000
Humidity	0.0168	0.0020	228.4228	0.0000
Year=2003	0.7436	0.1794	10.0601	0.0020
Year=2004	0.5791	0.2373	12.0416	0.0311
Year=2005	0.4525	0.2909	14.1493	0.1418
Year=2006	0.8099	0.3390	22.2157	0.0258
Year=2007	0.5344	0.4072	29.8599	0.1994
Year=2008	0.9944	0.4065	35.4881	0.0195
Year=2009	0.9620	0.4181	43.4492	0.0263
Year=2010	0.2832	0.4358	50.0631	0.5188
Year=2011	0.8921	0.4361	58.0866	0.0453
Year=2012	0.6931	0.4599	62.6088	0.1368
Year=2013	0.8726	0.4863	71.9257	0.0770
Year=2014	0.2123	0.4918	75.6820	0.6672
Year=2015	-0.0816	0.4785	80.6154	0.8650
Year=2016	0.2873	0.4910	87.6972	0.5600
Year=2017	-0.1119	0.5055	89.7039	0.8252
Year=2018	-0.8839	0.4975	89.6959	0.0790
Year=2019	-0.7619	0.5265	91.7696	0.1513
Year=2021	-0.5840	0.5294	104.3328	0.2725
Year=2022	-0.9325	0.4998	113.8678	0.0646
Year=2023	-0.6416	0.5071	125.9452	0.2082
Year=2024	-0.8379	0.5116	130.7013	0.1039
Year=2025	-0.7945	0.5329	134.3444	0.1383
Distance=2.95 Miles	0.2148	0.0837	253.1394	0.0109

Name	Estimated Variance
Athlete random intercept	10.8684
Random error	1.3667

Table 2: Model coefficient estimates for girls team

	Estimate	CRSE	Degrees of freedom	P-value
Intercept	22.0613	0.7621	124.8061	0.0000
Grade=10	-0.6292	0.1316	79.1687	0.0000
Grade=11	-0.9715	0.1769	94.0851	0.0000
Grade=12	-1.0341	0.2589	86.9462	0.0001
Percent improvement	-0.2084	0.0102	65.9980	0.0000
Temperature	0.0592	0.0076	111.6974	0.0000
Humidity	0.0115	0.0020	115.5790	0.0000
Year=2003	0.7014	0.2087	9.4813	0.0078
Year=2004	0.1393	0.3202	11.4688	0.6716
Year=2005	0.0953	0.3581	13.2086	0.7943
Year=2006	0.3529	0.4248	24.1021	0.4143
Year=2007	0.0046	0.5398	33.0717	0.9933
Year=2008	-0.0485	0.5765	43.7767	0.9333
Year=2009	0.1179	0.6343	56.1055	0.8532
Year=2010	0.0214	0.7588	66.6987	0.9776
Year=2011	0.0979	0.7198	75.4699	0.8922
Year=2012	0.4081	0.6725	76.8196	0.5458
Year=2013	1.1796	0.8613	81.4905	0.1746
Year=2014	0.9482	0.8073	79.6769	0.2437
Year=2015	0.4219	0.8104	86.1764	0.6039
Year=2016	0.7110	0.8224	83.2239	0.3897
Year=2017	-0.6143	0.7406	99.9008	0.4089
Year=2018	-1.2947	0.7575	100.3900	0.0905
Year=2019	-1.1588	0.8013	93.6456	0.1514
Year=2021	-1.5339	0.8446	96.5171	0.0725
Year=2022	-1.1606	0.9431	106.0604	0.2212
Year=2023	-1.0616	1.0114	102.0104	0.2964
Year=2024	-0.8938	0.9612	96.2607	0.3547
Year=2025	-1.1970	1.0735	90.4872	0.2678
Distance=2.95 Miles	0.3105	0.1087	123.9885	0.0050

Name	Estimated Variance
Athlete random intercept	16.4394
Random error	1.2151

6.4 Addressing model assumptions

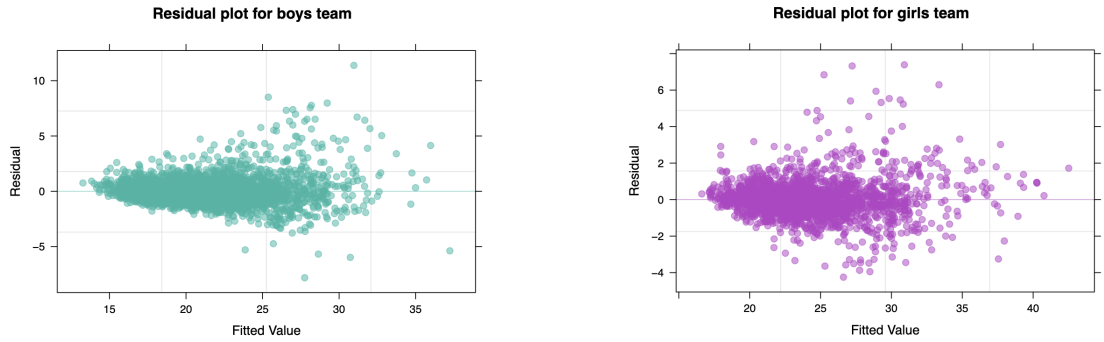

Figure 10: Residual versus fitted plots for both teams. Slower times tend to have larger residuals.

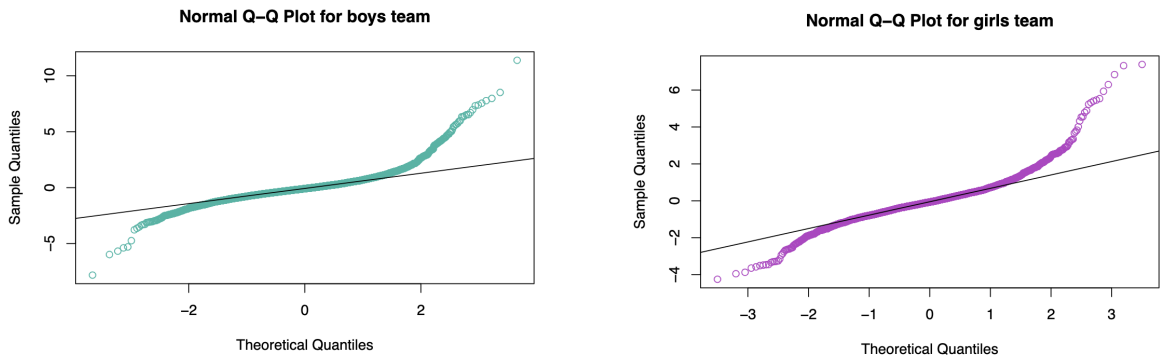

Figure 11: Q-Q plots for both teams. The center of the distribution is alright, but the tails diverge heavily.

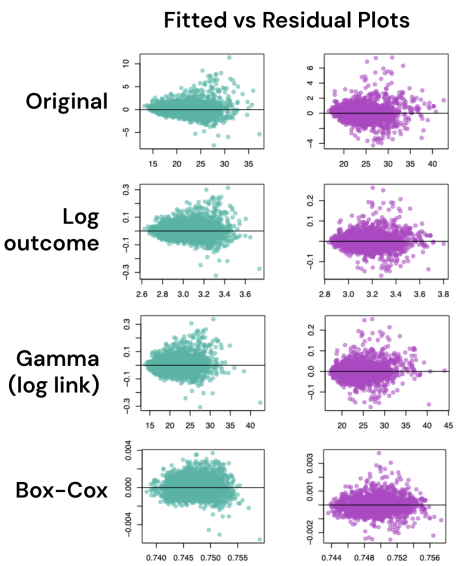

Figure 12: Residual plots for different model ideas, compared to the original.

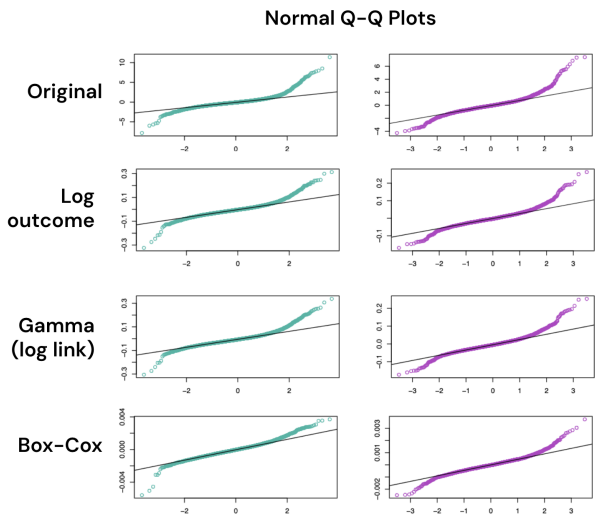

Figure 13: QQ plots for different model ideas, compared to the original.

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